EMERGENCY SERVICES SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture — the Operate-Through Sector

DBA-ESS: Emergency Services Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

Version 0.11 — DRAFT

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For review by engineers, regulators, public-safety leaders, and emergency-management and policy analysts responsible for the resilience of dispatch, response, and emergency-coordination capability.

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Emergency Services sector. It is the parent instrument for the DBA-ESS series and applies the DB → DBA → DBA-MA three-layer analytical progression to the functions on which public-safety response depends: emergency call-taking and dispatch, the response forces themselves, and emergency management and coordination.

1.2 What This Document Does Not Do

This document does not supersede emergency-management doctrine, public-safety regulation, or the National Incident Management System. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. For its consumer functions it does not re-scope the physical infrastructure response depends on: the communications, electric power, transportation access, and healthcare interface it relies on are governed by the DBA-COMM, DBA-EN, DBA-TRAN, and DBA-HPH series and inherited here.

1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The sector is organized by function — the answering and dispatch of emergency calls, the fielding of response forces, and the management and coordination of emergencies — and each function is placed by the public-safety consequence its failure produces, sub-indexed by response discipline and jurisdictional tier.

1.4 Series Placement — DBA-ESS as a Standalone Sector Series

DBA-ESS is a standalone sector series, a peer of the energy, water, chemical, communications, transportation, financial, healthcare, critical-manufacturing, and nuclear series. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 Emergency Services as the Operate-Through Sector

Emergency Services has a design-basis property no other sector shares as its defining feature: it must function precisely when the other sectors fail. It is the response layer for the whole critical-infrastructure framework, and its operational characteristic — stated plainly in the sector taxonomy — is that it must remain available during infrastructure failures, at all hours, with rapid response and multi-agency coordination. Where the other sectors design to withstand a degraded condition, Emergency Services designs to keep operating through it. Its design basis is therefore an operate-through requirement rather than a withstand requirement.

This reframes its dependencies. Emergency Services is a consumer sector on the model of the financial and healthcare sectors — it produces no physical commodity and rests on inherited

communications, power, transportation, and healthcare infrastructure — but with an inversion: it cannot assume those dependencies are available, because it is dispatched precisely when they are not. The communications-degradation baseline the corpus already defines (CDB) is, for this sector, not a design condition to be survived but the ordinary operating environment. Its call-taking-and-dispatch function is the operational demand-side dual of the DBA-COMM critical-services overlay: the 911, PSAP, and public-safety-radio infrastructure that DBA-COMM carries as a service is, here, the operating floor of the sector.

Section 2 — The Emergency Services Sector and Its Regulatory Context

2.1 A Distributed, Multi-Jurisdiction Sector

Emergency Services is the most jurisdictionally distributed sector in the framework — tens of thousands of independent law-enforcement, fire, and emergency-medical agencies at federal, state, and local levels, coordinated through doctrine rather than ownership. The Cybersecurity and Infrastructure Security Agency is the Sector Risk Management Agency; the Federal Emergency Management Agency provides the governing doctrine through the National Incident Management System and the National Response Framework; the Federal Communications Commission governs the Enhanced- and Next-Generation-911 requirements; and the National Fire Protection Association sets the consensus standards for fire and rescue. Operational authority, however, is overwhelmingly state and local.

2.2 Doctrine-and-Standard Inheritance by Function

The applicable authority varies by function: the dispatch function inherits the FCC 911 and Next-Generation-911 requirements and the FEMA communications doctrine; the response-forces function inherits the NFPA consensus standards and the state and local operating authorities and discipline-specific regulators; and the emergency-management function inherits the FEMA NIMS and National Response Framework doctrine that governs multi-agency coordination.

2.3 The Regulatory Gap

The sector is rich in operational doctrine and consensus standards but carries no engineering design basis stated in consequence terms — no statement of the response capability the sector must be demonstrated to sustain through a defined envelope of infrastructure failure. Preparedness is measured in plans, drills, and mutual-aid agreements; a design basis measured in demonstrated operate-through capability does not exist.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Discipline / Jurisdiction Sub-Index

The sector is decomposed into three function classes — public-safety answering and dispatch, response forces and stations, and emergency management and coordination. Each is sub-indexed by response discipline (law enforcement, fire and rescue, emergency medical services) and by jurisdictional tier (federal, state, local), so that the design basis load a given instance carries reflects both what it does and the population it is the responder of last resort for. Answering and dispatch and response forces are physical classes and emergency management and coordination is the coordination class, each on the author's decision of 6 October 2026, under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical).

3.2 Public-Safety Answering & Dispatch (AD)

The dispatch function covers the Public Safety Answering Points, the 911 and Next-Generation-911 systems, computer-aided dispatch, and the public-safety radio systems (P25 and the FirstNet public-safety broadband network). Its governing consequence is loss of the ability to receive an emergency call and direct a response — the point at which the public cannot summon help. It is the sector's tightest coupling to the communications sector and the operational counterpart of the DBA-COMM critical-services overlay. On the author's decisions of 6 October 2026, the boundary with the communications sector lies where the call and the caller's location enter the 911 system. Communications carries them that far: the originating networks and, in an enhanced 911 system, the 911 trunks, selective routing and location delivery that a carrier or 911 service provider operates. From that point inward the class is this one: the answering point's building, power, call-handling equipment and computer-aided dispatch, its backup answering point, and its telecommunicators. The Emergency Services IP Network of a Next-Generation 911 system, with the core services that route calls across it, and the agency-owned P25 radio systems belong here by ownership and consequence. FirstNet, a carrier network, belongs to DBA-COMM-AC-FC2 (*Access and Edge*), with the elevation obligation of Section 6. *Public-Safety Answering and Dispatch* §1.2 and §8.3 state the boundary in full. Where a function on the 911 authority's side is contracted, the contract test of Public-Safety Answering and Dispatch §8.3 decides its side: it is the authority's when the contract binds the provider to the class's design obligations for that function and the authority can enforce them, on the author's decision of 7 October 2026. On the author's confirmation of the same date, the core services of a Next-Generation 911 system sit with the Emergency Services IP Network on this side.

3.3 Response Forces & Stations (RF)

The response-forces function covers law enforcement, fire and rescue, and emergency medical services — their stations and apparatus and the field units they deploy. Its governing consequence is loss of response capability in a service area, and its distinctive feature is discipline heterogeneity: a police, fire, and EMS response are different capabilities with different design bases sharing a single dispatch and coordination layer. Its emergency-medical element is a boundary with the healthcare sector at the point of patient transfer and medical direction.

Response Forces and Stations develops the class, on the author's decisions of 6 October 2026. It is one class sub-indexed by discipline, typed physical, with staffing counted as on hand unless its arrival is assured; its governing measures are surviving response capacity and operate-through duration applied to units, with the mutual-aid map and its arrival time as the third parameter; and the loss of responders, to the hazard or to a deliberate second strike, is a consequence chain of its own. Command at a scene is exercised by the officers of the responding forces and passes to unified and multi-agency command, and with it to the coordination class, under the transfer-of-command rules the incident command system already sets; the reference to the incident-command structure at Section 3.4 is read with that transfer and does not redraw it. At the healthcare boundary, the field action of a medical crew and its design basis belong to this class and the clinical authority of the medical director to DBA-HPH. The pressure and flow of water for firefighting are a dependency on the water sector, and a water facility that is a fire district's only supply carries a tightened criterion for it (*Water Sector Framework* §8.3).

3.4 Emergency Management & Coordination (EMCo)

The emergency-management function covers the emergency operations centers, the incident-command structure, and the federal, state, and local emergency-management organizations. Its governing consequence is loss of the multi-agency coordination capability on which a large or compound incident depends — the function whose failure converts a manageable multi-agency response into an uncoordinated one. It is also the sector's coordination class, the locus at which the AI-Governance Design Basis is implemented: the Bright Line placement and Command-Broker obligation for machine-learning and generative-AI systems in dispatch prioritization, resource allocation, and incident decision support. On the author's decision of 6 October 2026, the origination of public warnings — the alerting authorities, their origination software, the federal aggregator and the outdoor warning sirens emergency management owns — belongs to this class, and the dissemination of alerts over the Emergency Alert System and Wireless Emergency Alerts belongs to DBA-COMM-AC-FC2 (*Access and Edge*). National Weather Service warnings are a federal input the class cites and does not own. *Emergency Management and Coordination* develops the class and states the Command Broker in full for the sector. On the author's decision of 6 October 2026, the Command Broker's independence in this sector is procedural, as *The Broker* §4.5 allows where the deploying authority's interest is aligned with correct placement. The decisions in which an official's other interests can run against the right act — an evacuation order delayed for its cost to a local economy, a warning softened to avoid alarm, an alert withheld for fear of being wrong — are recorded in *Emergency Management and Coordination* §6.6 rather than structurally separated. Recording them is part of the justification from the domain's incentive structure that *The Broker* §4.5 requires, and the holder's ability to refuse without answering to the person who gains remains the safeguard. This is the same decision the author made for the chemical and communications sectors.

† The AD and RF designators were ratified 2026-08-09; the dispatch-class code changed from DP to AD, where AD = Answering (the PSAP "Answering" Point) & Dispatch. On the author's decision of 6 October 2026, the

emergency-management class, ratified as DBA-ESS-EM-FC3, is re-designated DBA-ESS-EMCo: it is the sector's coordination class and carries no facility-class number, mirroring DBA-ES-CntlCo, DBA-CHEM-CorpCo and DBA-COMM-OpsCo. Emergency management carries the coordination role internally rather than as a separate coordination class, because coordination is a core operational function of this sector rather than an overlay.

3.5 The Operate-Through Consequence Metric

The corpus's consequence-duration metrics measure how long a degraded condition persists. The Emergency Services analog, proposed here for development, measures the sector from the other direction: the operate-through duration is the interval over which response capability must be sustained under a defined envelope of infrastructure degradation — loss of commercial power, of primary communications, of routine transportation access — before the response mission itself fails. It is the design-basis expression of the sector's defining requirement, and it is calibrated to the compound-failure scenario rather than to any single dependency. *Public-Safety Answering and Dispatch* states the metric for the answering function and, on the author's decision of 6 October 2026, pairs it with the transfer time to an alternate answering point.

On the author's decision of 11 October 2026, three statements returned by DBA-ESS-UC-1 (The Compound Failure the Sector Must Operate Through) complete the form of the metric; its values remain open (Section 9). First, the sector's operate-through duration for an area is the least of its class durations along the path from call to response: the answering point's, the units' and, for an incident too large for one agency, the operations center's. A call answered with no unit to send is not a response, and a class duration longer than the least buys nothing until the least is raised.

Second, a stock the classes share is counted once across the sector, in one record. The generators of answering points, radio sites and stations, and the tanks of the apparatus, draw on the same fuel on hand, and the telecommunicators, crews and operations-center staff come from the same population the event strikes. Each class counts what is on hand for its own endurance (Public-Safety Answering and Dispatch §3.2; Response Forces and Stations §3.2), and the sector's record keeps the same fuel or the same person from being counted twice, as the transportation series counts an alternative once across its campaign (Transportation Sector Framework §7.4).

Third, the metric has two forms. For the event, it is the duration over which the sector operates on what it holds. For the campaign that Section 7 makes part of the design duration under military-action conditions, when restoration is itself struck and the outage has no end date against which to count stocks, it is a balance: whether what the area can replace in each period, in fuel, crews, apparatus and stations, at least equals what the campaign consumes in it. Response Forces and Stations §5.6 already counts capacity this way under a campaign (*Repeat Strikes in Ukraine* F-6).

Section 4 — The Emergency Services Sector Design Basis

[Partly developed.] Section 4.2 states the principles the three classes found and the framework adopts for the sector. The design envelope by event category, with a consequence-severity inclusion criterion

consistent with the series, is still to be developed; the classes develop it ahead of the framework at their Section 4.

4.1 Design Basis Principles

The criterion for inclusion is public-safety consequence severity — potential for loss of the capability to receive emergency calls, to field a response, or to coordinate a large incident, at a scale and duration that leaves a population without effective response — not event likelihood. Candidate bounding events, to be developed: loss of a PSAP or the 911 system; loss of public-safety communications across a response area; station, apparatus, or response-force degradation; loss of an emergency operations center during an incident; and the compound infrastructure-failure scenario that the operate-through requirement is calibrated against.

4.2 Sector Principles Adopted From the Classes

The three class instruments each found, from their own reference events, principles that hold for the whole sector, and each asked the framework to state them once. On the author's decisions of 7 October 2026 the framework adopts five, and on the author's decision of 11 October 2026 two more, returned by the sector-level use case DBA-ESS-UC-1. Each is stated here and applied by the classes; none is restated in a class, and a class that applies one cites this section.

Conditionality. Any relaxation of a conservative class parameter must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. Each class lists the relaxations most likely in its domain (Public-Safety Answering and Dispatch §5.8; Response Forces and Stations §5.8; Emergency Management and Coordination §5.6).

Safeguards sized by reach. The safeguards on any element are sized by how far its failure or its wrong act reaches: the population a single alert reaches or the agencies a single order moves (Emergency Management and Coordination §3.1), the population whose calls pass through a single upstream element of the call path (Public-Safety Answering and Dispatch §3.4), and the area a single station or capability answers for (Response Forces and Stations §5.4). Concentration is recorded, in the alternate-routing map and the class maps, rather than made a parameter, and a concentration that is not recorded cannot be justified under the conditionality principle.

Time to authorize and time to correct inside the warning time. For each hazard whose warning time is measured in minutes, the time within which an above-line act can be authorized, and the time within which a wrong act can be corrected, must each be shorter than the warning time of the hazard named. The values are set by the jurisdiction and left open at class level (Emergency Management and Coordination §10.1; *False Missile Alert* F-5, F-6).

A human act at the point of origin. An act that originates an emergency call or a public warning requires a human act at the point of origin that presents what the act will do: the number to be called, the message and area to be warned. In Hawaii a statewide false alert was sent through

software whose one confirmation read the same for a drill and a warning and did not show the message (*False Missile Alert* F-3, F-4); in October 2016 a flood of 911 calls was originated by handsets that asked for no confirmation at all (*Call Flood* F-1). The principle binds the alerting authorities of the coordination class and, for the handset, the access class of the communications series, which states it for user terminals (Access and Edge, at its next revision).

Evidence held by the sector, not certified to it. Where a class depends on an element it does not own — a carrier office, a routing element, a hosted dispatch service, a water supply for firefighting — the class holds its own evidence of that element's protection on a cadence it sets, and does not rest on a certification made to a regulator; where a contract places the element on the sector's side, the contract binds the provider to supply that evidence (Public-Safety Answering and Dispatch §3.4, §8.3; *Derecho 911 Outages* F-5).

Residuals held where they are handed. A class may hand a declared residual to another class. Public-Safety Answering and Dispatch §5.7 hands the loss of call delivery across every carrier to the coordination class's instruction to go in person to staffed stations, and Response Forces and Stations §5.7 hands the region-wide loss of mutual aid to the coordination class and the federal response. A residual so handed is held only if the receiving class can show that it holds it within its own surviving capacity under the same design event, and in a compound event that capacity is being drawn down by the same cause. A residual the receiving class cannot show it holds is declared at the sector level, as a residual of this framework (DBA-ESS-UC-1 §4.2, §6). Two residuals are so declared on the author's decision of 11 October 2026. The loss of call delivery across every carrier, which Public-Safety Answering and Dispatch §5.7 hands to an instruction to go to staffed stations, and the region-wide loss of mutual aid, which Response Forces and Stations §5.7 hands to the coordination class and the federal response, are not held by the coordination class in the compound event, because the stations and the coordination they rely on are drawn down by the same cause (DBA-ESS-UC-1; DBA-ESS-UC-3). They are residuals of this framework: the population of an area that cannot summon help, and the gap until state and national resources arrive, for the design duration.

A heartbeat from every site. Each answering point, station and radio site reports its operating state to the operations center at a set interval whether or not anything has changed, so that the absence of a report is itself the alarm. A site that reports only on change sends nothing when it fails, and a silent site looks like a quiet one; the 2014 outage lasted six hours for that reason (*Multistate 911 Outage* F-1; Public-Safety Answering and Dispatch §7.2). The heartbeat travels on a channel that shares no carrier, power supply or network with what it reports on (CDB-4; *Ukraine Alerting* F-3). Its interval bounds the time to recognize a lost site, and so sets the recognition time counted inside the alternate-answering transfer time (Public-Safety Answering and Dispatch §3.2). It is authenticated, because an adversary who can counterfeit a heartbeat can hide a site it has taken down, and it is an indication FD-BL-D2 (Indication Integrity) governs. A missed heartbeat is read as a lost site until it is shown otherwise (DBA-ESS-UC-1 §6). On the author's decision of 11 October 2026 the principle extends to units: each unit reports its position

and status from the unit, authenticated there, over a path that does not pass through the dispatch system, so that the dispatch system's picture of the units can be checked against a source it does not produce (DBA-ESS-UC-2; Response Forces and Stations §3.2). The link from each detecting authority to an alerting authority carries a heartbeat on the same terms (DBA-ESS-UC-4).

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

Emergency services are adopting machine learning and generative AI in call triage and prioritization, in resource allocation and dispatch optimization, and in incident decision support. The Bright Line must be placed for any action of a component within scope of FD-LD §4.3 whose incorrect performance could misdirect or withhold a response at or above the severity threshold — a triage disposition, a dispatch decision, a resource-allocation call during a mass-casualty incident — with the architecturally enforced authorization of a qualified Command Broker. The sector's control systems — computer-aided dispatch, records management, emergency notification, and incident-command systems — are the deployment environment. Determinism decides scope under FD-LD §4.3; it does not decide placement. For dispatch, where an action must be taken in seconds, the author decided on 6 October 2026 that the authorization may be given in advance for a bounded class of actions, by the medical director for medical dispatch and by the responding agency for its response plans, with every action outside that class referred to a person in real time. This is advance discharge under FD-BL-D1 (*Mode of Discharge*), available only where its conditions are met, and *Public-Safety Answering and Dispatch* §6.4 develops it. On the author's decision of 11 October 2026, a deliberate attack in progress in an area makes the pre-authorized envelopes that serve it immutable, as a suspected compromise does, and while the attacker is unlocated no automated system dispatches to a reported deliberate attack (Public-Safety Answering and Dispatch §6.4; DBA-ESS-UC-3). Warning has the asymmetry of stop and proceed: an instruction that moves or releases people rests on a stronger basis than one that keeps them in protection (Emergency Management and Coordination §6.3; DBA-ESS-UC-4).

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Inward View — the Operate-Through Dependencies

Emergency Services depends on communications for its 911, radio, and dispatch links; on energy for facility operation and vehicle fueling; on transportation for the road access its response vehicles require; and on healthcare for patient transfer and medical direction. The inversion of Section 1.5 governs all of them: the sector is dispatched precisely when these are degraded, so each dependency is stated not as an availability assumption but as a degradation the sector must operate through. The dispatch function's dependence on the DBA-COMM critical-services

overlay is the sharpest of these, and the communications-degradation baseline (CDB) is its ordinary operating condition rather than a design excursion.

6.2 Outward View — the Mitigation Layer

The outward consequence of Emergency Services is unique: it is the mitigation layer for every other sector. A degraded response capability does not merely harm this sector; it amplifies the consequence of every other sector's failure, because the response that would have contained that failure is itself impaired. Emergency Services degradation is therefore a consequence multiplier across the entire framework, and its protection is a precondition for the effectiveness of every other sector's recovery.

6.3 Boundary with Healthcare

The emergency-medical element of the response-forces function is a boundary object with the healthcare sector at the point of patient transfer and medical direction: the pre-hospital response is DBA-ESS, the receiving facility and clinical direction are DBA-HPH, and the transfer is the shared interface both series must state consistently.

Section 7 — Military-Action Extension Applicability

[Skeleton.] Emergency Services is a second-wave military-action target: degrading dispatch, response, or coordination amplifies the consequence of a primary attack on another sector, and the deliberate timing of an attack to coincide with a degraded response is itself a military-action characteristic. The sector-level presumption toward MA applicability is strong for PSAPs, primary and backup emergency operations centers, and public-safety communications. Application remains a per-facility determination under the four-filter applicability gate of DBA-MA-AF (MA Applicability Framework) §2 and §3, in which any one filter satisfied brings a facility into scope. For emergency services facilities the strategic weight is driven by the population and the primary-target proximity the facility is the responder for. Two military-action principles, found independently by two classes from the war in Ukraine, are adopted for the sector on the author's decision of 7 October 2026. Under military-action conditions the design duration is a sustained campaign as well as an event: capacity is counted with the replacement of crews, apparatus, stations and the operators who send warnings, and a change to a system in use is a governed act exercised before it reaches the public (Response Forces and Stations §5; Emergency Management and Coordination §5.2; *Repeat Strikes in Ukraine* F-6; *Ukraine Alerting* F-6). And the design basis is not validated once and held: criteria — entry rules at a struck scene, alert levels, the indications that justify taking a system offline — are re-validated against observed adversary practice on a cadence the class sets, and a material change in that practice is a re-validation trigger (Response Forces and Stations §5.8; Emergency Management and Coordination §5.3; *Repeat Strikes in Ukraine* F-8; *Ukraine Alerting* F-5). Public-Safety Answering and Dispatch inherits both at its next revision.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments in the DBA-ESS series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework, and — as an operate-through consumer sector — inherit the physical design basis of their substrate from DBA-COMM, DBA-EN, DBA-TRAN, and DBA-HPH under the degradation assumption of Section 1.5. Each instrument adds the response-function-specific consequence chain and the operate-through recovery architecture.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-ESS series. All three classes — answering and dispatch, response forces, and the emergency-management coordination class — are at working-draft status, issued in October 2026. The inherited substrate is shown as governed by the communications, energy, transportation, and healthcare series. Designators were ratified 2026-08-09 and sub-indexed by response discipline and jurisdictional tier; the emergency-management designator was revised on 6 October 2026 (Section 3.4).

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory anchor
DBA-ESS-AD-FC1	Public-safety answering & dispatch — PSAPs, 911/E911/NG911, computer-aided dispatch, public-safety radio (P25, FirstNet)	operate-through duration and alternate-answering transfer time; operational demand-side dual of the DBA-COMM critical-services overlay	Working draft	FCC (E911/NG911); FEMA; state 911
DBA-ESS-RF-FC2	Response forces & stations — law enforcement, fire & rescue, EMS: stations, apparatus, and field units	surviving response capacity and operate-through duration applied to units; discipline sub-indexed (LE / fire / EMS); jurisdictional tier	Working draft	NFPA; state & local; discipline regulators
DBA-ESS-EMCo	Emergency management & coordination (coordination class) — EOCs, incident command, FEMA / state / local emergency management	authority reach and warning time; multi-agency coordination during infrastructure failure	Working draft	FEMA (NIMS, National Response Framework)
(inherited)	Substrate — communications (911/radio), electric power, transportation access, healthcare interface	governed by DBA-COMM / DBA-EN / DBA-TRAN / DBA-HPH	Governed elsewhere	see those series (operate-through inheritance)

Designators were ratified 2026-08-09 and confer no completed instrument. Facility-class numbers are sequential across the series. AD-FC1 and RF-FC2 are physical classes; EMCo is the coordination class and carries no FC number, mirroring DBA-ES-CntlCo, and was re-designated from EM-FC3 on 6 October 2026. Grounded in the CIKR_v16 sector map (Emergency Services Core Infrastructure and Operational Characteristics).

Section 9 — Development Status and Open Items

This is a working draft of the DBA-ESS sector framework. It establishes the standalone-series placement, the operate-through framing — the sector must function through the very failures the other sectors design to withstand — the function-primary decomposition (answering and dispatch, response forces, and the EMCo coordination class) with a discipline-and-jurisdiction

sub-index, the operate-through consequence metric, the inheritance of its substrate from DBA-COMM, DBA-EN, DBA-TRAN and DBA-HPH under a degradation assumption, and the mitigation-layer outward consequence. It is grounded in the CIKR_v16 sector map. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed; DBA-ESS-AD-FC1 (*Public-Safety Answering and Dispatch*), DBA-ESS-RF-FC2 (*Response Forces and Stations*) and DBA-ESS-EMCo (*Emergency Management and Coordination*) develop the class-level content ahead of them. All twelve reference-event instruments the three classes proposed were written on 7 October 2026, and each proposing class took in what they returned. On the author's triage ruling of 9 October 2026 two are withdrawn and ten are live, and under the determination of 9 October 2026 the ten are held by Project 11 (Section 9.1). Section 4.2 adopts five sector principles from the classes, and Section 7 two military-action principles. Open items at this version: develop the design-basis envelope of Section 4 by event category; set, with the energy and communications series, the design durations and the values each class leaves open at its Section 10.1; and set the values of the operate-through metric, whose form Section 3.5 now states from the compound-failure scenario, with the coverage floor and mutual-aid arrival time the response-forces class defines. The emergency-medical boundary with DBA-HPH was settled on 6 October 2026: the field action and its design basis are this sector's and clinical authority is healthcare's, for medical dispatch and medical crews alike (Section 3.3). Whether the communications series' critical-services overlay remains an overlay was settled on 6 October 2026: it stays an elevation overlay (*Emergency Management and Coordination* §8.3). The independence of the Command Broker for the conflicted decisions the coordination class records was settled the same day: it is procedural, and the decisions are recorded (Section 3.4). On the author's decision of 10 October 2026 the series will carry three use cases, one for each class and the first in the series, drawing on the material the reference-event instruments offered: a dispatch system corrupted rather than encrypted during a primary attack, for *Public-Safety Answering and Dispatch*; a second strike timed to the response, with misdirected calls, for *Response Forces and Stations*; and a counterfeit warning source and a false all-clear, for *Emergency Management and Coordination*. They are owed. On the author's decision of 11 October 2026 a fourth use case, at sector level, was issued ahead of them: DBA-ESS-UC-1 (The Compound Failure the Sector Must Operate Through) works the compound-failure scenario through all three classes and returns the form of the operate-through metric (Section 3.5) and two sector principles (Section 4.2). Under the series' numbering by order of issue the three class use cases take DBA-ESS-UC-2 to DBA-ESS-UC-4, in the order listed above. All four were issued on 11 October 2026: DBA-ESS-UC-2 (Dispatch Corrupted, Not Encrypted, During a Primary Attack), DBA-ESS-UC-3 (A Second Strike Timed to the Response) and DBA-ESS-UC-4 (A Counterfeit Warning Source and a False All-Clear), and the three classes took in what they returned at Public-Safety Answering and Dispatch v0.4, Response Forces and Stations v0.5 and Emergency Management and Coordination v0.6.

9.1 Conferred Reference Events

Under the author's decision of 6 October 2026 that every reference event a class proposes is conferred, this framework confers a reference-event designator on each of the events its classes propose. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event on the model of the water series' DBA-W-RE-1 (*Minnesota Water Campaign*) and DBA-W-RE-2 (*Oldsmar Water Treatment*). All twelve instruments were written on 7 October 2026. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are held by Project 11, DBA Reference Events; this framework keeps the conferral, and a designator's instrument is cited where Project 11 holds it. On the author's triage ruling of 9 October 2026 two of the twelve are withdrawn and their designators retired, and ten are live. The designators are numbered in the order the proposing classes were issued and, within each class, in the order the class lists its events. The number records conferral, not issue, and the instruments may be written in any order. Each designator is sector-primary and single: an event whose consequence reached other sectors is conferred here, in the sector that proposed it, and its reach is recorded in the register's impacted-sector column rather than in a second designator.

From *Public-Safety Answering and Dispatch*, five events are conferred: DBA-ESS-RE-1, the 911 outages after the derecho of 29 June 2012; DBA-ESS-RE-2, the multistate 911 outage of 9 and 10 April 2014; DBA-ESS-RE-3, the 911 call flood of October 2016, relabeled from the telephony denial-of-service attack on 911 centers on the author's decision of 7 October 2026, because "attack" overstated an event its instrument finds deliberate in its act and not established in its intent; DBA-ESS-RE-4, the ransomware attack on Baltimore's computer-aided dispatch of March 2018; and DBA-ESS-RE-5, the ransomware attack on Dallas's computer-aided dispatch of May 2023. The first two began on the communications side of the boundary of Section 3.2. They are conferred here because their consequence was the loss of 911 answering, and the communications sector is recorded in their impacted-sector column. Hurricane Katrina, which the class cites as an illustration, is not conferred. The Nashville bombing of December 2020 and the AT&T outage of February 2024, which reached 911 service, are already DBA-COMM-RE-4 and DBA-COMM-RE-7, and the class cites them.

From *Emergency Management and Coordination*, four events are conferred: DBA-ESS-RE-6, the false ballistic missile alert in Hawaii of 13 January 2018; DBA-ESS-RE-7, the warning failure in the Lahaina wildfire of 8 August 2023; DBA-ESS-RE-8, the failure of coordination in Hurricane Katrina in 2005, which *Public-Safety Answering and Dispatch* cited as an illustration for its 911 losses; and DBA-ESS-RE-9, Ukraine's air-raid alerting from February 2022, the series' first reference event that records an answer rather than a failure.

From *Response Forces and Stations*, three events are conferred: DBA-ESS-RE-10, Hurricane Harvey in August 2017, in which stations and apparatus were lost to flooding and crews were short; DBA-ESS-RE-11, the shooting at the Route 91 Harvest festival in Las Vegas on 1 October 2017, a mass-casualty surge across the patient-transfer boundary; and DBA-ESS-RE-12, the repeat strikes on rescue crews in Ukraine from February 2022, the series' first reference event in

which the responders are the target. The class also cites the explosion at West, Texas, which is DBA-CHEM-RE-5, and Hurricane Katrina, which is DBA-ESS-RE-8, and the Lahaina wildfire, conferred as DBA-ESS-RE-7 and withdrawn (below), which the class now cites as an illustration (*Response Forces and Stations* §2.3).

Two conferrals are withdrawn on the author's triage ruling of 9 October 2026, recorded in Project 0's notice of that day. DBA-ESS-RE-7, the warning failure in the Lahaina wildfire, is withdrawn and its designator retired. Its F-1, that the alerting authority could not see whether its warnings were delivered, is carried into DBA-ESS-RE-9 (*Air-Raid Alerting in Ukraine, 2022 to 2026*) v0.2 as its preface, so that the gap and Ukraine's practice of knowing a channel is down and saying so stay paired; *Emergency Management and Coordination* keeps the recommendations it took in from DBA-ESS-RE-7 at its v0.3, with a provenance line at its v0.5. DBA-ESS-RE-10, Hurricane Harvey, is withdrawn and its designator retired, with nothing carried; *Response Forces and Stations* records the withdrawal at its v0.4 and keeps Harvey as an illustration cited to its sources. Both instruments stay with Project 9, in its archive, and do not move to Project 11.

The ten live instruments, held by Project 11, are: *The 911 Outages After the Derecho of June 2012* (DBA-ESS-RE-1); *The Multistate 911 Outage of April 2014* (DBA-ESS-RE-2); *The 911 Call Flood of October 2016* (DBA-ESS-RE-3); *The Ransomware Attack on Baltimore's Computer-Aided Dispatch, March 2018* (DBA-ESS-RE-4); *The Ransomware Attack on the City of Dallas, May 2023* (DBA-ESS-RE-5); *The False Ballistic Missile Alert in Hawaii* (DBA-ESS-RE-6); *The Failure of Coordination in Hurricane Katrina* (DBA-ESS-RE-8); *Air-Raid Alerting in Ukraine, 2022 to 2026* (DBA-ESS-RE-9); *The Route 91 Harvest Shooting of October 2017* (DBA-ESS-RE-11); and *Repeat Strikes on Rescue Crews in Ukraine, 2022 to 2026* (DBA-ESS-RE-12). The two withdrawn are *The Warning Failure in the Lahaina Wildfire* (DBA-ESS-RE-7) and *Hurricane Harvey and the Houston Response of August 2017* (DBA-ESS-RE-10). Of the ten, four were non-adversarial; four were deliberate acts that did not meet the military-action threshold (RE-3, RE-4, RE-5 and RE-11); two are military-action events (RE-9 and RE-12); and RE-9 is the series' one event that records an answer rather than a failure. They are cited in this document by short title: *Derecho 911 Outages*, *Call Flood*, *False Missile Alert*, *Ukraine Alerting* and *Repeat Strikes in Ukraine*.

References

[To be completed.] Governing and inherited instruments: FD-BL (*The Bright Line*) and its determinations FD-BL-D1 (*Mode of Discharge*) and FD-BL-D2 (*Indication Integrity*); FD-LD (*Layer Decomposition and Determinism*); FD-LS (*Local Sovereignty*); FD-BR (*The Broker*); FD-EV (*The Envelope*); the DBA-MA parent framework; DBA-MA-CDB (*Military Action Communications Design Basis*), the communications design basis, whose degraded baseline is this sector's ordinary operating condition; DBA-MA-AF (*MA Applicability Framework*); CB-PAE (*Pre-Approved Action Envelope*), in the Command Broker family; DBA-COMM (*Communications Sector Framework*), whose critical-services overlay covers 911, the Emergency Alert System and the priority services, and DBA-COMM-AC-FC2

(*Access and Edge*); DBA-EN (facility power and fuel); DBA-TRAN (response access); DBA-HPH (patient-transfer and medical-control boundary); DBA-W (*Water Sector Framework*, fire-flow dependency, §8.3); FD-SBL (*The Surveillance Bright Line*), for the liberty consequence of law-enforcement actions, with version state in FD-SBL-REG; DBA-W-RE-1 (*Minnesota Water Campaign*) and DBA-W-RE-2 (*Oldsmar Water Treatment*), the models for the reference-event instruments of Section 9.1. The Foundational Definitions and the DBA-MA instruments are cited without a version token; FD-REG (*Governing Stack Register*) is authoritative for the Foundational Definitions' version state, and CB-REG (*Command Broker Family Register*) for the Command Broker family's. Class instruments: DBA-ESS-AD-FC1 (*Public-Safety Answering and Dispatch*), DBA-ESS-RF-FC2 (*Response Forces and Stations*) and DBA-ESS-EMCo (*Emergency Management and Coordination*). Model coordination class: DBA-COMM-OpsCo (*Operations and Coordination*). Regulatory and doctrine: CISA (SRMA); FEMA (NIMS, National Response Framework); FCC (E911 / NG911); NFPA; state and local authorities. Grounding: CIKR_v16 sector map (Emergency Services Sector). Reference-event instruments: the ten live instruments, held by Project 11, and the two withdrawn, DBA-ESS-RE-7 and DBA-ESS-RE-10, kept by Project 9, listed with their titles at Section 9.1. Use cases: DBA-ESS-UC-1 (The Compound Failure the Sector Must Operate Through) v0.2; DBA-ESS-UC-2, DBA-ESS-UC-3 and DBA-ESS-UC-4, v0.1.

Revision History

0.11 October 2026 — T. Roxey. Takes in the findings the three class use cases returned to the framework, on the author's decisions of 11 October 2026. Section 4.2 extends the heartbeat to units and to the link from each detecting authority (DBA-ESS-UC-2, DBA-ESS-UC-4), and declares two sector-level residuals, the loss of call delivery across every carrier and the region-wide loss of mutual aid, which the coordination class cannot hold in the compound event (DBA-ESS-UC-1, DBA-ESS-UC-3). Section 5 freezes the pre-authorized envelopes of an area during a deliberate attack and states the asymmetry of warning (DBA-ESS-UC-3, DBA-ESS-UC-4). Section 9 records the issue of DBA-ESS-UC-2 to DBA-ESS-UC-4 and the class revisions that took in their findings: Public-Safety Answering and Dispatch v0.4, Response Forces and Stations v0.5 and Emergency Management and Coordination v0.6. Revision-history entries are left as records.

0.10 October 2026 — T. Roxey. Takes in the findings of DBA-ESS-UC-1 (*The Compound Failure the Sector Must Operate Through*) v0.2, the series' first use case, on the author's decision of 11 October 2026. Section 3.5 states the form of the operate-through metric: the sector's duration for an area is the least of the class durations along the path from call to response, a stock the classes share is counted once across the sector, and the metric has an event form and a campaign form. Section 4.2 adopts two further sector principles: a residual handed between classes is held only where the receiving class can show it holds it, and every answering point, station and radio site sends an authenticated heartbeat on an independent channel. Section 9 records the use case, issued ahead of the three class use cases, which take DBA-ESS-UC-2 to DBA-ESS-UC-4, and restates the metric's open item as its values. The classes apply the two new principles at their next revisions. Revision-history entries are left as records.

0.9 October 2026 — T. Roxey. Records the triage of the series' reference events and their move to Project 11. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*) the ten live instruments are held by Project 11, and Section 9.1 and the References say so; the conferral stays here. On the author's triage ruling of 9 October 2026, Section 9.1 withdraws DBA-ESS-RE-7 (the Lahaina wildfire) and DBA-ESS-RE-10 (Hurricane Harvey) and retires both designators: DBA-ESS-RE-7's F-1 is carried into DBA-ESS-RE-9 (*Air-Raid Alerting in Ukraine, 2022 to 2026*) v0.2 as its preface, and *Emergency Management and Coordination* keeps the recommendations it took in from DBA-ESS-RE-7, with a provenance line at its v0.5; nothing is carried from DBA-ESS-RE-10, which *Response Forces and Stations* records as withdrawn at its v0.4. Section 9 records the author's decision of 10 October 2026 that the series will carry three use cases, one for each class, and that they are owed. An independent check of the revision against the notes, the instruments and the classes raised 2 items: the pointer for the Lahaina illustration at Section 9.1, corrected to *Response Forces and Stations* §2.3, and the pending notice to Project 11, given the same day. Project 11 is given notice of this revision. Revision-history entries are left as records.

0.8 October 2026 — T. Roxey. On the author's decisions of 7 October 2026, Section 4.2 adopts five sector principles the classes found: conditionality, safeguards sized by reach with concentration recorded, the time to authorize and to correct inside the warning time, a human act at the point of origin of an emergency call or warning (DBA-ESS-RE-3 R-6), and evidence of elements the sector does not own held by the sector rather than certified to it; Section 4 is marked partly developed. Section 7 adopts two military-action principles found by two classes from the war in Ukraine: the design duration is a campaign, and criteria are re-validated against observed adversary practice. Section 3.2 records the contract test and the Next-Generation 911 confirmation. Section 9 records all twelve reference-event instruments as written and taken in, and restates the open items. Section 9.1 relabels DBA-ESS-RE-3 as the 911 call flood and lists the twelve instruments. The running header and footer are brought to the house form. Revision-history entries are left as records.

0.7 October 2026 — T. Roxey. The register enters DBA-ESS-RF-FC2 (Response Forces and Stations) at working-draft status, completing the sector's three classes, and types it physical. Section 3.3 records the author's decisions of 6 October 2026 on the class: its measures, its responder chain, the passing of command to the coordination class under the incident command system's own transfer rules, the medical-control boundary with DBA-HPH, and the fire-flow dependency on the water sector. Section 9 restates the open items: the response-forces instrument is written and the emergency-medical boundary is settled. Section 9.1 confers DBA-ESS-RE-10 through DBA-ESS-RE-12 on the three reference events the class proposed. References name the class, the water framework and FD-SBL. Revision-history entries are left as records.

0.6 October 2026 — T. Roxey. Section 3.4 records the author's decision of 6 October 2026 that the Command Broker's independence in this sector is procedural, and that the decisions in which an official's other interests can run against the right act, named in Emergency Management and

Coordination §6.6, are recorded rather than structurally separated, as in the chemical and communications sectors. Section 9 restates the open items accordingly. Emergency Management and Coordination §6.6 and §10.4 still describe the question as open and conform at their next revision. Revision-history entries are left as records.

0.5 October 2026 — T. Roxey. The register enters DBA-ESS-EMCo (Emergency Management and Coordination) at working-draft status. Section 3.4 places the origination of public warnings, including the federal aggregator and emergency management's outdoor sirens, in that class and their dissemination in DBA-COMM-AC-FC2, with National Weather Service warnings a cited input, on the author's decision of 6 October 2026. Section 9 restates the open items: the question of the critical-services overlay is settled, the overlay staying an elevation overlay, and the independence of the Command Broker for the conflicted decisions the coordination class records is added. Section 9.1 confers DBA-ESS-RE-6 through DBA-ESS-RE-9 on the four reference events the coordination class proposed, and states the numbering rule for more than one class. References name the coordination class and its model. Revision-history entries are left as records.

0.4 October 2026 — T. Roxey. Section 3.1 types the classes under the author's determination of 30 September 2026: answering and dispatch is physical and emergency management is the coordination class, each on the author's decision of 6 October 2026, and the type of the response-forces class waits on its instrument. Section 3.4 and the register re-designate DBA-ESS-EM-FC3 as DBA-ESS-EMCo, a coordination class without a facility-class number, on the author's decision of the same date. Section 3.2 states the boundary with the communications sector where the call and the caller's location enter the 911 system, with the Emergency Services IP Network, its core services and agency-owned P25 radio systems in this sector and FirstNet in DBA-COMM-AC-FC2, on the author's decisions of the same date. Section 3.5 records the pairing of the operate-through metric with alternate-answering transfer time. Section 5 records that dispatch authorization may be given in advance under FD-BL-D1, on the author's decision of the same date. Section 7 applies the four-filter gate of DBA-MA-AF in place of the two-factor test. The register enters DBA-ESS-AD-FC1 (Public-Safety Answering and Dispatch) at working-draft status. Section 9 restates the open items, and Section 9.1 confers DBA-ESS-RE-1 through DBA-ESS-RE-5 on the five reference events the class proposed. References give designations with titles in place of the bare designation of the parent framework. Cover date corrected. Revision-history entries are left as records.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.2 August 2026 — T. Roxey. Designators ratified; dispatch-class code changed DP → AD (Answering & Dispatch); AD / RF / EM confirmed; three-class, no-separate-coordinator structure (EM absorbs coordination) ratified.

0.1 August 2026 — T. Roxey. Initial skeleton, the first sector grounded in the CIKR_v16 sector map. Established DBA-ESS as a standalone sector series with the operate-through framing (it must function through the failures other sectors design to withstand; the CDB communications-degradation baseline is its ordinary operating condition; its dispatch function is the operational demand-side dual of the DBA-COMM critical-services overlay). Adopted the function-primary decomposition (Public-Safety Answering & Dispatch / Response Forces & Stations / Emergency Management & Coordination) with a discipline-and-jurisdiction sub-index; proposed the operate-through consequence metric; stated the mitigation-layer outward consequence and the DBA-HPH patient-transfer boundary. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.